

a. After freezing of the magma ocean, Earth was rapidly rotating and oblate

Compositional crust:

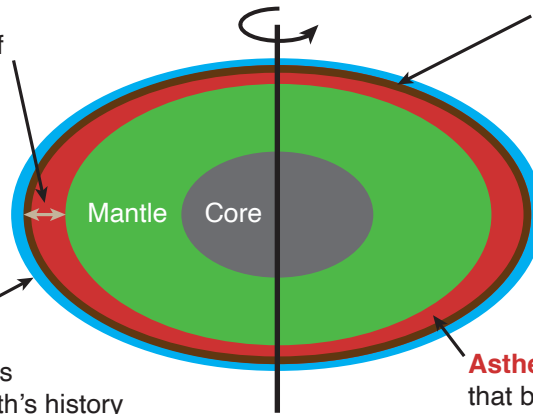
Formed by extraction of melt from the mantle. Lower gravity leads to thicker crust closer to the equator

Lithospheric crust:

Behaves as a solid over relevant timescale. Initially uniformly thick.

Surface ocean: Oceans condensed early in Earth's history while it was still rapidly rotating

Asthenospheric crust: Crust that behaves as a fluid over the relevant timescale. Initially, most of the crust is asthenospheric



b. Earth's shape changed substantially during early lunar recession, driving key processes

Both latitudinal and longitudinal extension and associated volcanism at high latitudes

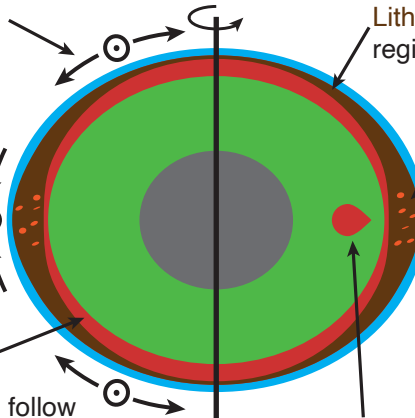
Convergent tectonics in the equatorial regions

Lithosphere thinned in polar regions and thickened in equator

Crustal melting and metamorphism associated with tectonics

Asthenospheric crust flows to follow an equipotential and becomes thicker in the polar regions

Possible delamination of lower crust



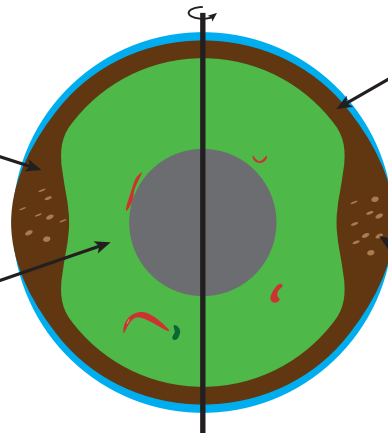
c. After tens of millions of years Earth was slowly rotating with minor oblateness

Crustal thickness and density variation inherited from despinning process

Chemical and isotopic heterogeneity recorded in the crust and mantle

Whole crust is **lithospheric**

More evolved, potentially felsic, rocks



Decreasing rotation rate with time